Mapping of Hay-Harvesting Grasslands Using Harmonized Landsat Sentinel-2 Time Series and Deep Learning in Temperate Steppe

Wei Lu, Yunfeng Hu, Batunacun, Jia Liu, and Hao Li

***Abstract*—Natural hay-harvesting grasslands are essential for sustaining livestock through winters in semi-arid temperate steppes. However, limited historical records and the absence of systematic spatial data hinder regional management. Moreover, although threshold-based and machine learning approaches have demonstrated effectiveness in monitoring mowing in Western Europe, their applicability in Eurasian semi-arid steppes remains uncertain. Therefore, we aimed to develop an approach for mapping natural hay-harvesting grasslands in semi-arid temperate steppes. Specifically, based on optical satellite time-series data, we employed a time-series classification architecture, light inception with boosting technique (LITE), to identify hay-harvesting grasslands. Results of our experiments, conducted in a typical region in Inner Mongolia, demonstrated that our approach can generate high-quality hay-harvesting grassland maps, with the testing accuracy of 92.06% ($F1$-score) and 90.65% [Overall accuracy (OA)]. Furthermore, we applied the gradient-weighted class activation mapping (Grad-CAM) technique to interpret the decision-making process of the deep learning model. Our findings showed that LITE concentrated intensively on time steps surrounding mowing dates, underscoring its superiority in mapping the geospatial extent of hay-harvesting grasslands and its potential for extracting temporal information related to hay harvesting. This study offers valuable insights for developing fine-scale hay-harvesting grassland inventories in semi-arid temperate steppe, an area long overlooked in existing research, thereby supporting the sustainable management of grassland resources, and promoting the advancement of smart herding practices.**

***Index Terms*—Deep learning, hay-harvesting grassland mapping, optical remote sensing, temperate steppe, time series.**

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I. Introduction

As ONE of the largest terrestrial ecosystems, grasslands are widely distributed across all continents except Antarctica, covering approximately 30% the terrestrial surface [1]. In China, natural grasslands account for 40% of the national land area [2]. China's grassland ecosystems, predominantly distributed across northern regions, serve dual critical functions as both ecological safeguards and economic resources. These vast grasslands constitute vital ecological barriers that mitigate sand and dust storm incursions into China's interior regions. Simultaneously, they play an indispensable role in ensuring national food security by supporting livestock production systems that provide meat and dairy products for domestic consumption [3], [4]. Geographically, China's grassland ecosystems are primarily situated within temperate continental climate zones characterized by semi-arid to arid conditions, making them ecologically fragile, highly susceptible to disturbances, and less capable of self-regulation [5], [6], [7]. Thus, reasonable management decisions on grassland utilization are essential to regulating the balance between ecological values and economic values.

China's grasslands experienced intensive grazing utilization during earlier developmental periods. The widespread overgrazing that intensified throughout the 1990s precipitated significant grassland degradation, and the instability of forage supplies was gradually exposed. This degradation was particularly manifested in the critical shortfall of cold-season forage production. Influenced by the temperate continental climate, livestock is lack of grass during cold winter and dry spring [8]. A significant shortage of fodder can lead to serious economic losses for herders [9]. To alleviate the issue of seasonal imbalance between forage supplies and livestock needs, hay-harvesting grasslands emerge as an effective complementary management measure. Herders mow fresh grass in later summer and early autumn, and store dry hay to prepare for cold-season forage. Thus, hay-harvesting grasslands are the strategic resources in semi-arid areas for ensuring the safety

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of livestock through winter and for emergency relief after snowstorms or droughts. Spatial information about the hay-harvesting grasslands can help to make adapted management decisions and carry out targeted monitoring of the total area of hay-harvesting grasslands, their distribution pattern, the amount of their production, or the degree of degradation [10], [11], [12]. However, systematic geospatial records of grassland utilization in northern China remain remarkably limited, primarily due to long-standing neglect in documenting historical spatial data by both management authorities and herders. This scarcity hinders understanding of how many hay-harvesting grasslands are needed to produce sufficient fodder to meet livestock's winter needs, how to define degradation thresholds for hay-harvesting grasslands, and other related issues. The establishment of a comprehensive mowing event dataset is a fundamental requirement for constructing an accurate inventory of China's semi-arid rangelands. Although field-based parcel surveys can generate such data at local scales, automated techniques, particularly remote sensing methods, are indispensable for regional-scale implementation.

Existing studies of mowing detection in Europe have primarily focused on intensively managed meadows and pastures within semi-natural permanent grassland systems, where mowing practices are widespread [13], [14], [15]. Although highly successful, the adaptability and transferability of methods developed for European permanent grasslands to temperate steppes remain uncertain and require further investigation. Benefiting from the temperate maritime climate, meadows in Europe can be mown multiple times per year, making the cycles of growth, harvesting, and regrowth clearly observable in satellite imagery. Hay-harvesting grasslands are less intensively managed in semi-arid temperate steppes, growing naturally without fertilization. Grass growth in these areas exhibits a single peak season. Thus, the grass is mown once a year at most in semi-arid temperate steppes, which is much different from the intensively managed meadows in Europe. Moreover, in northern China, mowing typically occurs from late August to September, coinciding with the natural senescence period of grasslands. This temporal overlap presents additional challenges for accurately detecting mowing events, especially those occurring in late September. Consequently, further research is required to develop methods specifically tailored to mapping hay-harvesting grasslands in temperate steppes.

Current methods using remote sensing imagery to monitor mowing can be grouped into three categories: visual interpretation-based, threshold-based [15], [16], and model-based [14], [17]. Two Chinese studies [18], [19] sought to map hay-harvesting grasslands in 2010s in northern China by visually interpreting Landsat TM images, relying on the differences in appearance characteristics observed before and after mowing events. The temporal changes of vegetation indexes (VIs), such as normalized difference vegetation index (NDVI), enhanced vegetation index (EVI), are the proxy indicators to identify the abrupt changes caused by mowing. The threshold-based method detects mowing via the decrease of VIs after the event or the difference between the ideal growth curve and actual growth curve [20], [21]. For instance, Griffith et al. [13] first computed the ideal growth NDVI profile determined by peaks and local maxima points and then set a threshold of residuals between the fit and observed values to identify potential mowing events in Germany. Schwieder et al. [16] defined a set of rules which include the threshold of the difference between observed EVI values and interpolated ones, the threshold of residuals of consecutive observations, and the minimum time intervals between mowing events. These threshold-based methods are easy to perform, though, lack of adaptability because other factors such as clouds and heavily grazing can also cause abrupt decreases in NDVI, and therefore, thresholds vary across regions and climates. Model-based methods typically feed VI profiles or other auxiliary features into machine learning or deep learning models to determine whether the curve indicates mowing events. Lobert et al. [14] and Holtgrave et al. [17] transformed mowing mapping into a binary classification task, split time series in a year into equally spaced time sequences, and then fed the sequences into the models to identify whether they covered a mowing event or not. For computing mowing frequency in Germany, Lange et al. [22] trained a CNN model to directly predict the number of mowing events, which is a multiclass classification task. However, the deep learning approaches are typically regarded as black box, since they are data-driven and lack of explicit logical rules.

Time-series remote sensing data was typically used to monitor mowing [17], [20], [22]. Medium-resolution optical remote sensing imagery, such as Sentinel-2 and Landsat imagery, has become the predominant data source for grassland monitoring due to its high-frequency revisit, adequate spatial resolution, large swath, and broad coverage [16]. Quite a few studies utilized coarse-resolution imagery, such as MODIS [21], which lacks detail, or high-resolution imagery, such as RapidEye [23], which has a narrow swath and is not freely available for large-scale applications. There are lot of data gaps in optical images due to weather conditions, while synthetic-aperture radar (SAR) data are not affected by weather. Therefore, some studies have utilized SAR data to monitor grassland mowing, inspired by the principle that changes in vegetation height result in changes in backscattering signals and coherence values [14], [24], [25]. However, the commonly used SAR product, Sentinel-1 A/B, has spatial and temporal resolutions that are inconsistent with those of optical datasets, necessitating additional processing for integration. Merely, interpolation algorithms are effective to fill data gap in recent studies in the optical images [14], [17].

Overall, the development of spatial datasets for hay-harvesting grasslands in semi-arid temperate steppes has long been neglected and is increasingly urgent for supporting regional development. Additionally, the applicability of threshold-based and machine learning methods, developed for intensively managed meadows in Europe, remains uncertain for natural hay-harvesting grasslands in temperate steppes due to substantial regional differences. Therefore, the objective of this study is to initially propose an effective approach for mapping hay-harvesting grasslands in temperate steppe. The

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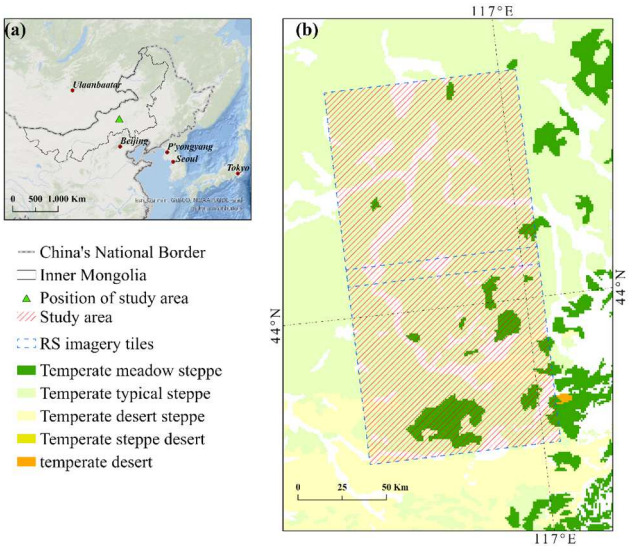


Fig. 1. Illustration of our study site. (a) Location of our study site (presented in WGS 1984 Web Mercator). (b) Grassland types of our study area (presented in Krasovsky_1940 Alberts) [27].

significance of this study will be demonstrated by addressing these questions

1) Can natural hay-harvesting grasslands in the semi-arid temperate steppe of northern China be effectively mapped using time series remote sensing data?
2) How well does a deep learning-based time series classification model perform in this context?
3) Which time steps are prioritized by deep learning models when identifying hay-harvesting grasslands?

This study can provide valuable insights for management of agro-pastoral systems in Eurasian semi-arid temperate steppe through addressing these questions.

II. Study Site and Data

A. Study Site

Our study site, covering an area of 23 057.03 km^2, is located in Inner Mongolia, China [Fig. 1(a)], which belongs to temperate continental climate with the annual average temperature of 1 °C–2 °C and annual precipitation of 100–350 mm [26]. Our study area encompasses predominantly temperate typical steppe, with scattered occurrences of temperate meadow steppe and temperate desert steppe [Fig. 1(b)] [27]. Among these three types, temperate typical steppe accounts for 88.8%, while temperate meadow steppe and temperate desert steppe account for 7.5% and 3.7%, respectively, in our study area. Grasslands within the study area mainly consist of hay-harvesting grasslands and grazing grasslands, which are utilized specifically for mowing and grazing, respectively. Grassland mowing events only occur in the temperate meadow steppe and temperate typical steppe, with a maximum frequency of once per year, due to the semi-arid climate. According to regional regulation on protection and utilization of natural mowing grassland, grassland owners in the study area are permitted to mow only after August 15. The hay-harvesting grasslands grow naturally without livestock activities before or after the mowing events in a year.

B. Image Data

This study primarily utilized Harmonized Landsat and Sentinel-2 (HLS) data, a product jointly developed by NASA and the USGS by integrating imagery from Landsat-8 and Sentinel-2 [28]. The combining algorithms include atmospheric correction, geographic registration, spatial resampling, bidirectional reflectance distribution function normalization, and bandpass adjustment. The HLS offers a revisit frequency of two and three days and a spatial resolution of 30 m. HLS dataset includes two products: S30 (from Sentinel-2) and L30 (from Landsat-8), both of which were used to construct the time series data cube. We acquired HLS data for the 2023 growing season, spanning from March 1 to October 30, covering the designated study area. Two HLS tiles [Fig. 1(c)], named "T50TMP" and "T50TMQ", totally covering 22 660.56 km^2, were used with the projection of WGS 84/UTM zone 50 N (EPSG:32650). Totally, 350 images with quality bands were acquired for the two tiles. We used the red band and near-infrared band to calculate the NDVI [29]. We also acquired European Space Agency (ESA) WorldCover 10 m 2021 dataset [30] to extract grassland extent. The ESA landcover data were reprojected by WGS 84/UTM zone 50 N and resampled at 30-m resolution. The land cover category assigned to each 30-m resolution pixel represents the majority class derived from its nine constituent 10-m resolution pixels. Subsequent classification was conducted within the grassland extent.

C. Reference Data

In this study, we acquire reference data for model training and testing in two ways, visual interpretation, and a filed survey. For the training dataset, we visually identified 569 parcels including 363 hay-harvesting grassland parcels and 206 non-hay-harvesting grassland parcels based on the time-series RGB images from HLS data, and Planet's daily satellite imagery that has high spatial resolution of 1 m provided by Planet Labs Inc [31]. A mowing event leads to a rapid reduction of the ground vegetation within a few days. This change is reflected in RGB imagery as a sudden decline in greenness, which contrasts in color with the surrounding non-hay-harvesting grasslands. Also, the hay meadows show a more compact and uniform texture than grazed pastures where livestock graze as they go along, so that the grass has high variability in height and density. It is reliable to visually identify the locations of hay-harvesting grassland from RGB imagery. Fig. 2 shows some examples of the visually interpreted parcels. For the testing dataset, samples were collected during a field survey from September 1, 2024 to September 7, 2024. A total of 195 field plots were recorded using handheld GPS devices and digital cameras, with classifications confirmed through direct observation and interviews with local herders. Subsequently, plot categories were validated by analyzing long-term time series data from HLS and Planet imagery captured in 2023. After this verification, 183 field samples exhibiting consistent long-term patterns and reliable field identification were retained as the final test dataset, including 122 hay-harvesting grassland plots and

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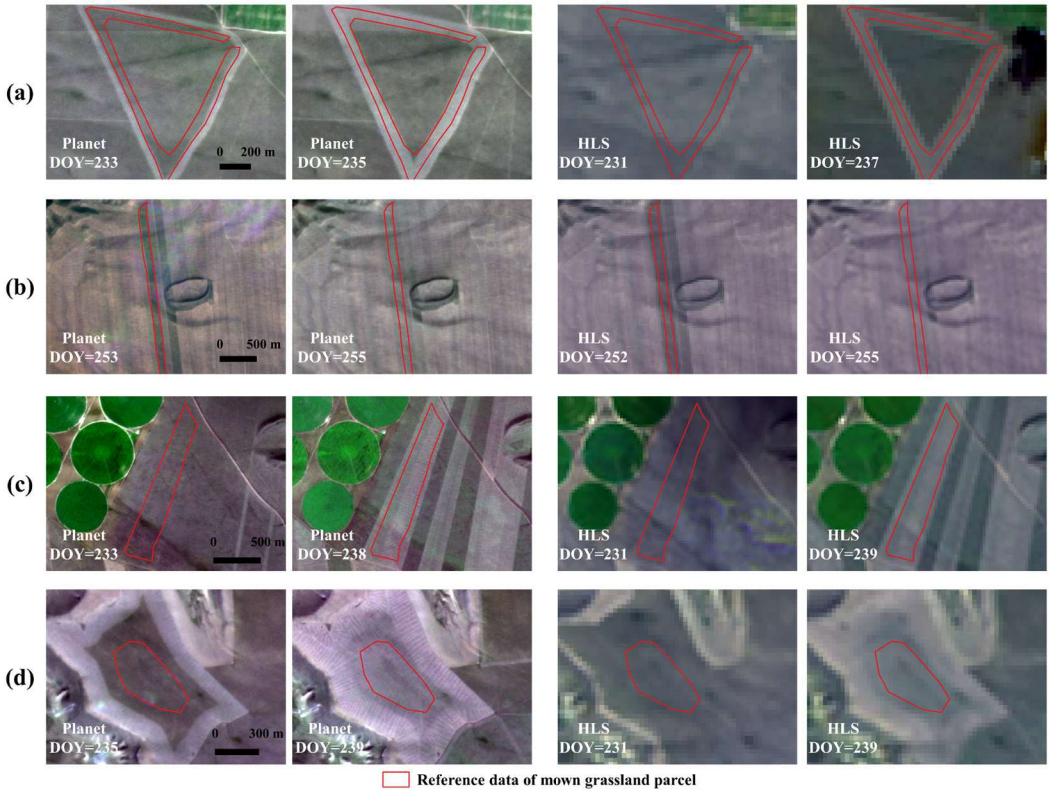


Fig. 2. Examples of reference mowing grassland parcels. Each row denotes one example. The first and second columns are Planet RGB images before and after the mowing event, respectively. The third and fourth columns are HLS RGB images before and after the mowing event, respectively. (a)–(d) Represent different examples, respectively. DOY = the day of year.

TABLE I
DESCRIPTION OF DATA SOURCES USED IN OUR STUDY

Name	Bands	Spatial resolution	Frequency of revisit	Usage	Source
HLS	Red, NIR	30-meter	2~3 days	For model training, testing, and predicting	https://hls.gsfc.nasa.gov/ [28]
Planet	Red, Green, Blue	1-meter	~1 day(s)	For interpreting the categories of reference samples	https://www.planet.com/[31]
ESA Landcover	-	10-meter	1 year	For generating grassland mask	https://esa-worldcover.org/en/data-access[30]

61 non-hay-harvesting grassland plots. More description of the criteria for the selection of samples was presented in the supplemental material S1.

We applied a 30-m inner buffer for each reference parcel to avoid mixed pixels at the boundaries. All the reference parcels were converted into raster files with a projection of WGS 84/UTM zone 50 N and a spatial resolution of 30 m. The reference data are binary, of which "0" denotes non-hay-harvesting grassland and "1" denotes hay-harvesting grassland. Totally, there are 171 267 pixels in the training dataset, including 66 839 hay-harvesting grassland pixels and 104 428 non-hay-harvesting grassland pixels; and 25 619 pixels in the testing dataset, including 15 096 hay-harvesting grassland pixels and 10 523 non-hay-harvesting grassland pixels. All datasets and their usage are listed in Table I.

III. METHOD

Our hay-harvesting grassland mapping method mainly contained three parts: time series generation, model implementation, and model interpretation (Fig. 3). As for the workflow of our approach, we first generated a seamless data cube of NDVI via the interpolation and filtering algorithm (Section III-A). Then, the time series data were input into a lightweight deep learning-based model, light inception with boosting technique (LITE) [32] to separate hay-harvesting grasslands from non-hay-harvesting grasslands (Section III-B). To find out which time steps of the series data the model focuses most intensively on, we introduced gradient-weighted class activation mapping (Grad-CAM) into assessing the importance scores [33], [34] (Section III-C), which can reflect the potential relationship between model attention

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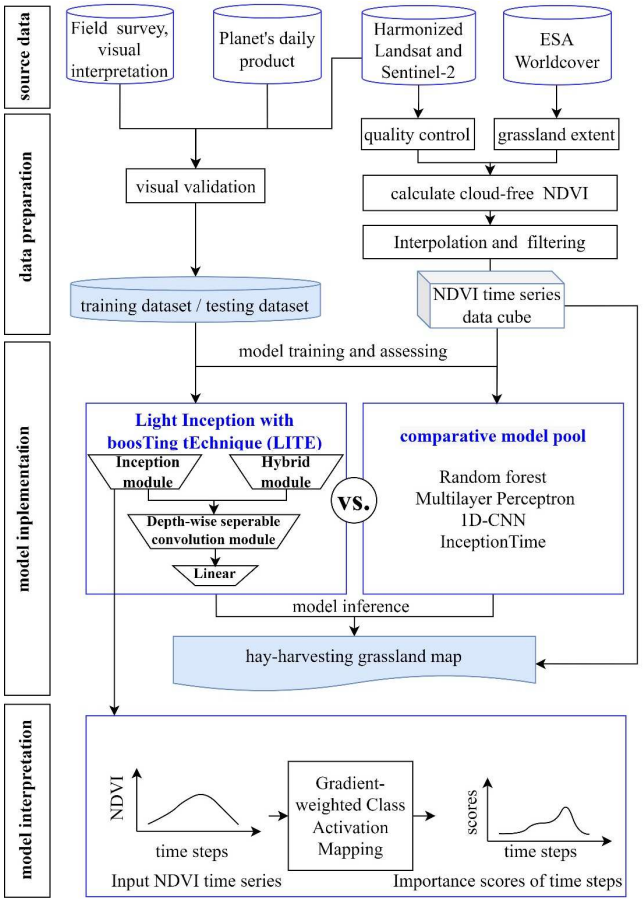


Fig. 3. Framework of this study.

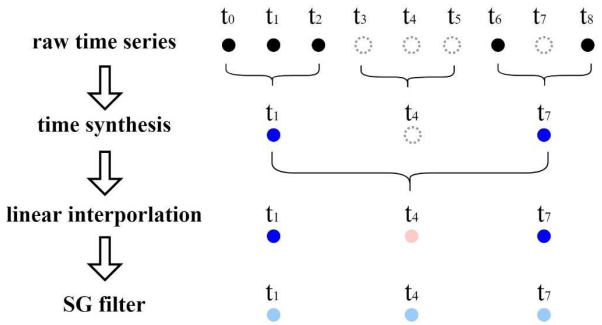


Fig. 4. Sketch of the procedure of time series generation. The black solid dot denotes the observed value; the black hollow dot denotes the missing value; the indigo dot means synthesis value; the pink dot means the interpolation value; the light blue dot denotes filtered value.

and observed mowing events, providing insights into the interpretability of deep learning models.

A. Time Series Generation

The concept underlying our study was based on pixel-level time series classification. We used the interpolation algorithm and time filtering method to generate NDVI time series. Fig. 4 is a sketch of the procedure of time series generation.

Quality control on NDVI time series data is necessary to enhance performance [35], [36]. First, we used the quality

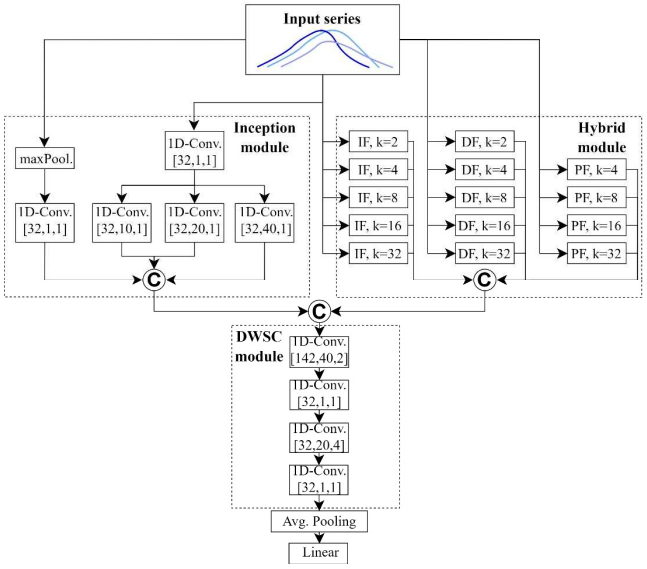


Fig. 5. Architecture of LITE. maxPool denotes the maximum pooling layer; Avg Pooling denotes the average pooling layer; 1-D-Conv means the 1-D convolution, and [32, 1, 1] means 32 filters, kernel size of 1 and dilation of 1; IF denotes the increasing DF; denotes the decreasing DF; PF denotes the PF; $k = 2$ denotes the filter has a kernel size of 2.

bands of HLS data to filter out cloud pixels, shadow pixels, pixels around clouds, and snow pixels. Then, based on the raw cloud-free NDVI time series, we used a sliding window with the size of n days to generate a synthesized NDVI time series, via a maximum filter. The corresponding synthesized NDVI value was defined as "nan" if no effective values were within consecutive n days at a certain pixel location. The time synthesis procedure was used to adjust raw irregular time series to regular time series, as the classification model we used was based on the Euclidean feature space. After time synthesis, we computed the final NDVI time series by using the linear interpolation to fill gaps and a Savitky–Golay (SG) filter to smooth the data [37], [38], [39]. We set the filter order as 2 and the filter length of 7. The impact of gap handling methods will be discussed later. Also, the time interval parameter n was set as 3 in our experiment and the suitability of this parameter will be discussed later.

B. Model Implementation

We used the light inception with boosting technique (LITE) model [32] to distinguish hay-harvesting grasslands and non-hay-harvesting grasslands. Adapted from InceptionTime [40], [41], [42], though, the LITE had much fewer parameters while keeping performance. Fig. 5 shows the architecture of LITE.

The LITE contains three main modules, that is, the inception module, the hybrid module, and the depth-wise separable convolution (DWSC) module. The Inception module of LITE consists of one bottleneck layer and three-parallel 1-D convolutional layers. The original LITE removes the bottleneck layer [32]. Considering the bottleneck layer can reduce the number of parameters [42], while improving performance at the same time, we kept the bottleneck layer in this study. The bottleneck layer contains one 1-D convolutional layer with the

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kernel size of 1, one 1-D max pooling layer with the kernel size of 3 and one 1-D convolutional layer with kernel size of 1. The kernel sizes of three 1-D convolution layers in the Inception module are 40, 20, and 10.

The hybrid module includes three kinds of handcrafted filters whose parameters are fixed and not learned during model training [43]: the increasing detection filter (IF), the decreasing detection filter (DF), and the peak detection filter (PF). The capability of the handcrafted filters to capture specific patterns in time series data has been demonstrated in recent studies. Specifically, the IF, defined as follows, was used to detect the increasing trend among time steps:

$$w_{I_i} = (-1)^{(i+1)}, \quad i \in \{0, 1, 2, \ldots, k-1\} \tag{1}$$

where w_{I_i} denotes the ith weight value in the IF; k denotes the kernel size of the filter. When $k = 4$, the IF is [−1, 1, −1, 1]. We used five IFs with kernel size of 2, 4, 6, 8, 16, 32.

The DF, defined as (2), was used to detect the decreasing trend among time steps

$$w_{D_i} = (-1)^{(i)}, \quad i \in \{0, 1, 2, \ldots, k-1\} \tag{2}$$

where w_{D_i} denotes the ith weight value in the DF. When $k = 4$, the DF is [1, −1, 1, −1]. We used five DFs with kernel size of 2, 4, 8, 16, 32.

The PF was used to detect the peak (first increasing and then decreasing), defined by dividing the filter into six parts and using the positive squared parabolic function and the negative squared parabolic function to calculate each part as follows:

$$w_{P_i} = \begin{cases} -x^2, & x = \frac{i\%(v//4)+1}{v//4},\ 0 \le i < v//4 \\ -x^2, & x = 1 - \frac{i\%(v//4)}{k//4},\ v//4 \le i < v//2 \\ 2x^2, & x = \frac{i\%(v//4)+1}{k//4},\ v//2 \le i < (3v)//4 \\ 2x^2, & x = 1 - \frac{i\%(v//4)}{v//4},\ (3v)//4 \le i < v \\ -x^2 & x = \frac{i\%(v//4)+1}{v//4},\ v \le i < (5v)//4 \\ -x^2 & x = 1 - \frac{i\%(v//4)}{v//4},\ i \ge (5v)//4 \end{cases} \tag{3}$$

where w_{P_i} denotes the ith weight value in the PF; v is a super parameter; the kernel size k of PF is $v + v//2$; the operation % here means taking the remainder, and the operation // means taking the quotient. When $v = 4$, k is 6, and the PF is [−1, −1, 2, 2, −1, −1]. We used four PFs with the super parameter of 4, 8, 16, 32, and kernel size of 6, 12, 24, 28, correspondingly. The output feature vectors of the Inception module and the aforementioned hybrid module are concatenated and fed into the DWSC modules.

A LITE model has two DWSC modules. Each DWSC module has one depthwise convolutional layer first and then one pointwise convolutional layer [44], [45]. For the first DWSC module, the depthwise convolutional layer has 142 filters and the kernel size of 40. For the second DWSC module, the depthwise convolutional layer has 142 filters and the kernel size of 20. The pointwise convolutional layers of two DWSC modules both have 32 filters and the kernel size of 1. The LITE finally has a global average pooling layer and a densely connected layer to produce the output.

After model training, we used testing datasets to evaluate the performance. We calculated four measures including precision, recall, overall accuracy (OA), and $F1$-score, which are commonly used for accuracy assessment [46], [47]. Precision denotes the proportion of correctly classified hay-harvesting grassland pixels from those were output by models, recall denotes the fraction of hay-harvesting grassland pixels that were correctly classified versus the reference. $F1$-score is a harmonic average of precision and recall. OA is a general measure index, calculated based on the confusion matrix. We defined the category of hay-harvesting grassland as "positive" and the category of non-hay-harvesting grassland as "negative."

Finally, we used the trained model to generate the hay-harvesting grassland map of our study site. The NDVI time series images were merged into one Geotiff file via layer stacking in date order. The synthesized file was fed into the trained model pixel by pixel. To provide a clear hay-harvesting grassland map, we used a median filter to reduce the potential salt-and-pepper noise [48], [49].

C. Model Interpretation

To improve the interpretability of our method, we introduced Grad-CAM [33], [34], [50] to visualize how the model makes a decision and highlight the most important time period that contributes to output. We defined an input time series as $S \in \mathbb{R}^{T \times D_{\text{in}}}$, where T is the number of time steps and D_{in} is the number of features. D_{in} is 1 in this study because we only used one vegetation index, NDVI. We used $A \in \mathbb{R}^{T \times D_{\text{in}}}$, where D_{out} is the number of filters, to represent the activation matrix of the target layer, and used $(\partial y_k)/(\partial A_{i,d})$, where y_k denotes the model output (before the softmax classifier) of the kth class, to represent the gradient value of the ith time step and the dth filter. Then, the class-specific gradients with respect to activation matrix A via backpropagation can be used to quantify the importance scores q_d of time steps for the dth filter, computed as below

$$q_d = \frac{1}{T} \sum_{i=1}^{T} \frac{\partial y_k}{\partial A_{i,d}}. \tag{4}$$

Finally, the Grad-CAM values from the target layer is calculated as follows:

$$L_{\text{Grad-CAM}} = \text{ReLU}\left(\sum_{d=1}^{D} q_d A^d \right) \tag{5}$$

where ReLU is the rectified linear unit, A^d is the activation tensor of the dth filter. We defined the Inception module as the target layer, since LITE extracts high-level features from time series data via the Inception module.

D. Experimental Settings

For model comparison, we investigated the performance of another four models, including a common-used machine learning classifier (random forest (RF) [51], [52]) and three artificial neural networks (ANNs) (multilayer perceptron (MLP) [53],

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TABLE II
RESULTS (%) OF THREE TRAILS (MEAN ± STD.) FOR DIFFERENT MODELS

model	Precision	Recall	F1	OA
RF	88.71±0.11	80.93±0.34	84.64±0.20	82.69±0.20
MLP	91.15±0.42	75.50±2.01	82.58±1.19	81.24±1.06
1D-CNN	87.66±1.97	87.24±2.02	87.42±0.21	85.13±0.48
InceptionTime	86.39±1.20	93.90±0.91	89.98±0.36	87.86±0.33
LITE	92.04±0.57	92.09±0.87	92.06±0.44	90.65±0.44

[54], 1-D convolutional neural network (1D-CNN) [14], [55], [56] and InceptionTime [40], [41], [42]). We ran experiments for three trials and computed the mean and standard deviation of each measure index.

We implemented the RF model with Scikit-learn package [57] and ANN models with PyTorch framework [58] based on Python. For the RF model, we set the number of trees to 200 while keeping the other parameters at their default values. For the ANN models, we set the batch size to 4096, and each model was trained for 100 epochs. The learning rate was $1e^{-3}$, and the weight decay was set to $1e^{-5}$.

IV. RESULTS

A. Overall Results

Accuracy assessment results generated by different models were presented in Table II. The comparison results showed that LITE outperformed all other models, while MLP exhibited the poorest performance. LITE obtained the highest OA of 92.06%, exceeding the other four models by 2.79%~9.41%, and the highest $F1$-score of 90.65%, exceeding the others by 2.08%~9.48%. In general, models such as 1D-CNN, InceptionTime, LITE, which accounted for the temporal dependency embedding the time series data, outperformed models like RF and MLP, which disregarded the temporal order.

In terms of precision and recall, LITE achieved over 90% in both metrics. InceptionTime recorded a slightly higher recall than LITE by 0.81%, but its precision was 5.56% lower than LITE, indicating a higher rate of false positives. MLP had the second highest precision but the lowest recall, suggesting that it generated a large number of omissions. 1D-CNN demonstrated a balanced performance between precision and recall, albeit with moderate scores on both metrics. Notably, LITE was the only model to achieve both high precision and recall within the testing set, demonstrating its best capability for handling errors and omissions.

We constructed a supplemental dataset, totally covering 270.95 km^2, to fascinate qualitative assessment within our study region [Fig. 6(a) and (b)]. We visually interpreted a pixelwise hay-harvesting map based on Planet's time series images for the supplemental dataset [Fig. 6(c)]. The results predicted by models were presented in Fig. 6(f)–(h), which also demonstrated that models including LITE and InceptionTime produced fewer errors or omissions than the other models.

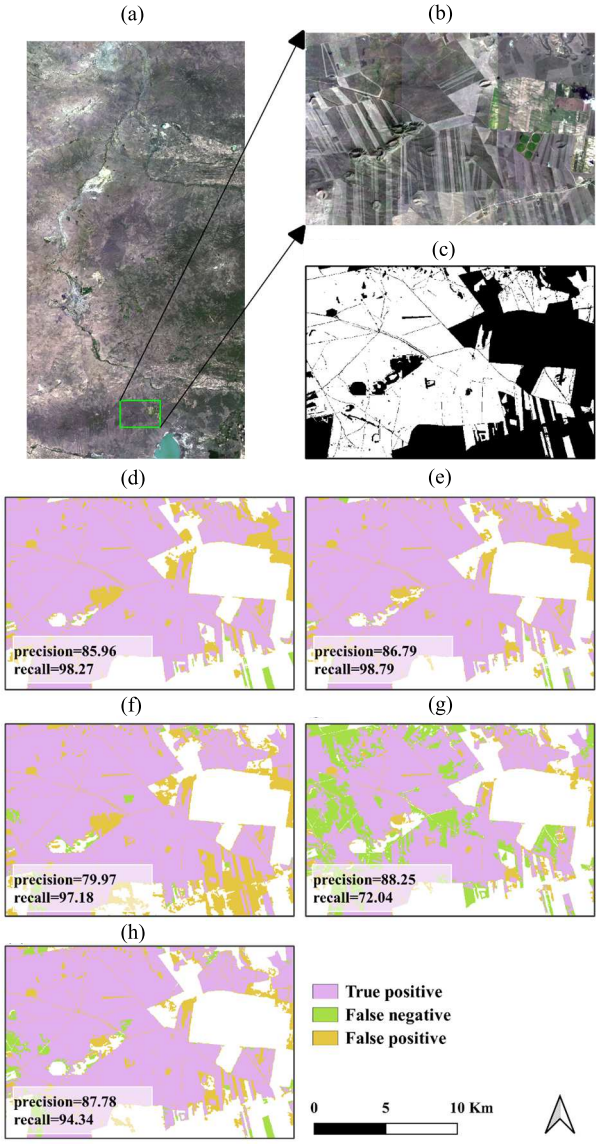


Fig. 6. Comparison of results generated by different models for the supplemental dataset. (a) Location of the supplemental dataset (base map: HLS image). (b) Planet's image (DOY = 241). (c) Reference. (d) LITE. (e) InceptionTime. (f) 1D-CNN. (g) MLP. (h) RF.

B. Results for Different Grassland Types

In order to assess the model performance within different grassland types, we divided the samples in our testing dataset into two groups according to the grassland type maps released by Tang [27], that is, temperate typical steppe and temperate meadow steppe. As for quantitative analysis, Fig. 7 presented the precision and recall values of different models within temperate typical steppe and temperate meadow steppe. All models performed better in temperate meadow steppe than in temperate typical steppe. Specifically, within the samples

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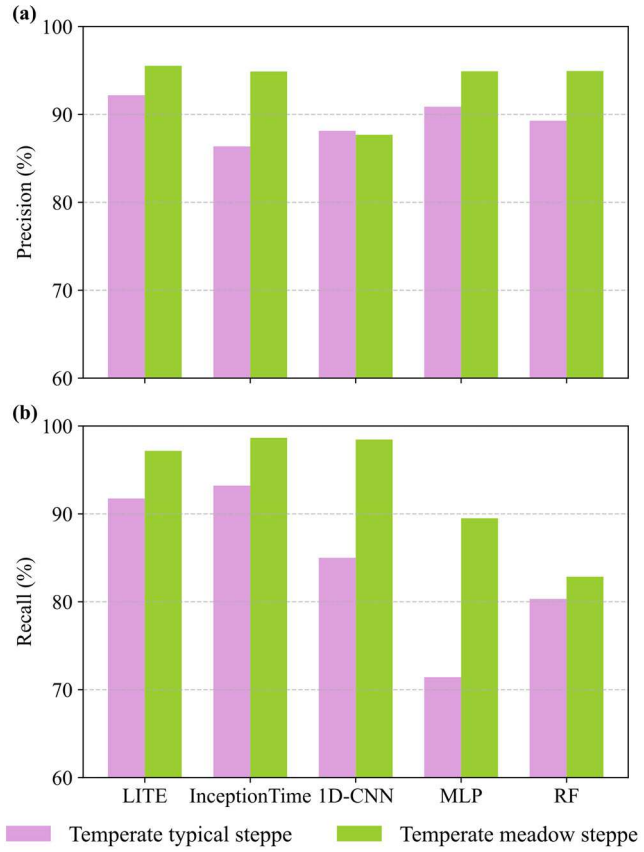


Fig. 7. Model accuracies under different grassland types.

belonging to temperate typical steppe, models including RF, MLP, and 1D-CNN produced more omissions than the other models, resulting in lower recall values. For 1D-CNN, a high recall was achieved in the temperate meadow steppe, but it also yielded the lowest precision. Compared to InceptionTime, LITE achieved precision and recall above 90% for both grassland types, demonstrating its robustness across different grassland types.

From qualitative aspect, we selected three regions within our study area, representing temperate typical steppe, temperate meadow steppe, and temperate desert steppe, respectively. The HLS RGB time series were given in Figs. 8–10 for visual comparison. In Fig. 8, the hay-harvesting grassland boundaries delineated by LITE are as clear and well defined as those observed in the RGB time series, whereas the boundaries produced by RF appear fragmented. Hay-harvesting grasslands are widely distributed in temperate meadow steppes, as shown in RGB time series of Fig. 9. LITE produced patchy hay-harvesting grasslands, while the other models, especially for RF and MLP generated omissions. Grasslands in temperate desert steppes are unsuitable for harvesting because of the lack of water conditions. As shown in Fig. 10, LITE detected hay-harvesting grasslands outside the temperate desert steppes, while MLP and 1D-CNN generated lots of errors within the temperate desert steppes. The qualitative comparison also confirmed that models, particularly RF, MLP, and 1D-CNN, resulted in more omissions or errors for different grassland types.

C. Results Under Different Gap Handling Methods

In this section, we evaluated the impact of different gap handling methods on model performance. The first method, nan-to-zero, directly fills gaps in the cloud-free three-day time series with zeros. The second method, linear interpolation, uses a linear interpolation algorithm to fill missing values. The third method combines linear interpolation with SG filtering, as described in Section III-A. Fig. 11 presented the evaluating results within the testing dataset.

Filling gaps using the nan-to-zero method led to substantial classification errors or omissions across all models, indicating its limited effectiveness. For example, LITE can produced a high median value of recall when using nan-to-zero approach, though it also generated a low precision score at the same time. Zero-filled gaps introduced additional noise into the time series NDVI data, thereby impairing the models' learning performance.

Applying linear interpolation significantly enhanced the classification performance of models such as LITE and InceptionTime. In particular, the precision scores of both models improved markedly after missing values in the time series data were filled using linear interpolation. We also observed that applying SG filtering after linear interpolation led to slight improvements in both precision and recall scores for LITE.

In overall, effective gap filling is a critical step in time series classification. While linear interpolation proves adequate for the majority of deep learning models, subsequent SG filtering offers additional performance gains.

D. Model Performance on Sequence Data Across Different Time Intervals

We set the time intervals of the input NDVI time series to 1, 3, 5, and 7 days, and compared the model performances using time series data with different temporal resolutions (Fig. 12). The center of the ellipse close to the upper right corner indicates that the current model has high precision and recall. A smaller ellipse corresponds to a lower standard deviation of the current model's performance. According to Fig. 12 LITE obtained good recall values and precision values across all different time intervals. Especially, when the time interval was set to three days, LITE generated both precision and recall values over 90%. RF and MLP were prone to produce higher precision values than recall values under different time intervals, indicating large number of omissions. 1D-CNN generated moderate precision and recall values under 90% across all the time intervals. Moreover, LITE demonstrated greater stability across different time intervals, as evidenced by the smaller size of its representative ellipses, especially in comparison to InceptionTime's.

E. Model Interpretation Based on Grad-CAM

In this section, we provided the visual interpretation of how input data at different time steps contribute to model decisions. LITE extracts deep features via an inception module. Thus, we set the inception module as the target layer when performing Grad-CAM to calculate the importance scores across all time steps. To facilitate comparison, we also applied Grad-CAM to

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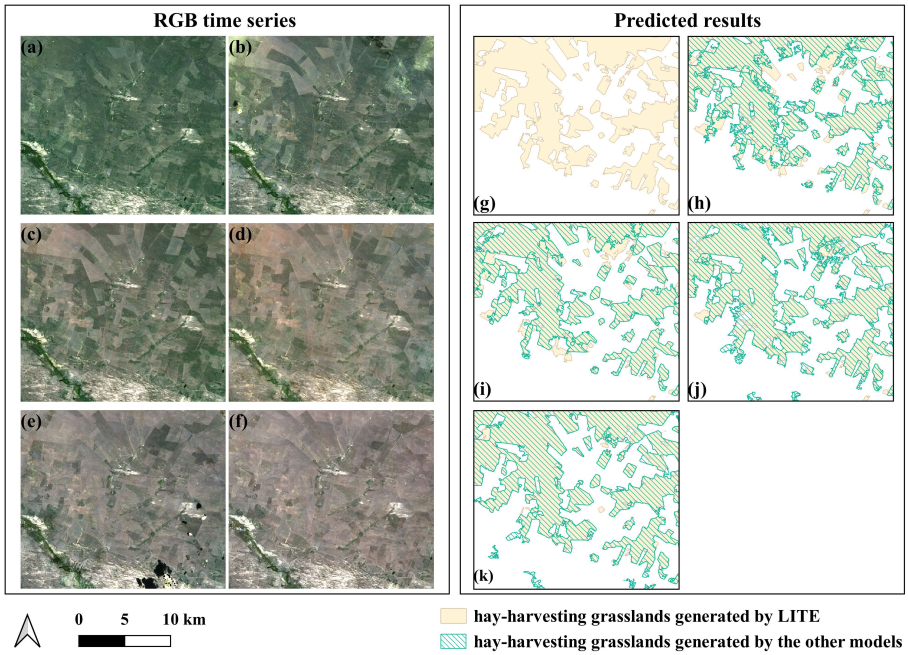


Fig. 8. Comparison of results generated by different models in a selected region within temperate typical steppe. (a) DOY = 239. (b) DOY = 247. (c) DOY = 252. (d) DOY = 257. (e) DOY = 263. (f) DOY = 271. (g) LITE. (h) LITE versus RF. (i) LITE versus MLP. (j) LITE versus 1D-CNN. (k) LITE versus InceptionTime.

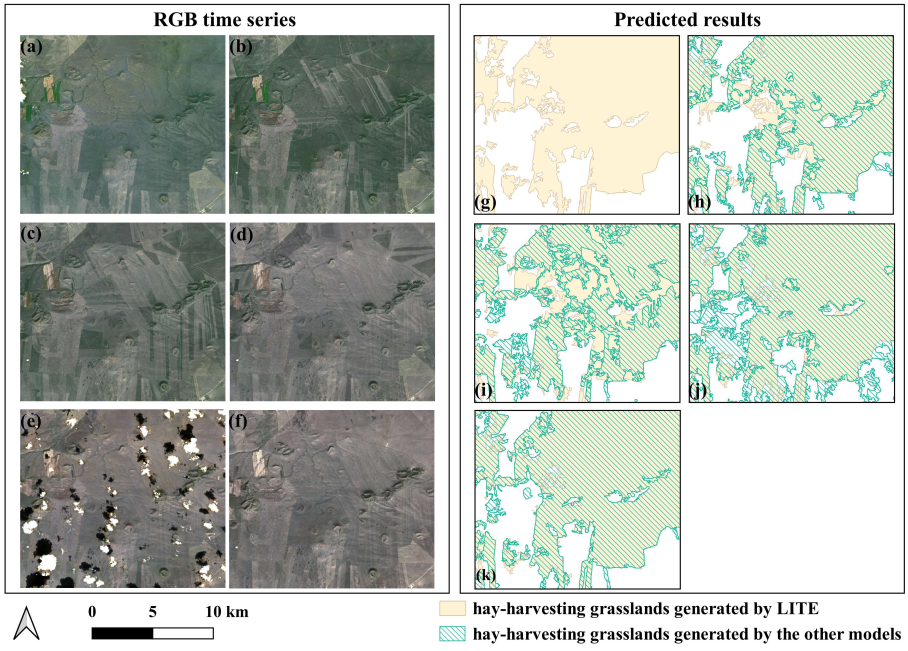


Fig. 9. Comparison of results generated by different models in a selected region within temperate meadow steppe. (a) DOY = 227. (b) DOY = 239. (c) DOY = 247. (d) DOY = 255. (e) DOY = 262. (f) DOY = 271. (g) LITE. (h) LITE versus RF. (i) LITE versus MLP. (j) LITE versus 1D-CNN. (k) LITE versus InceptionTime.

InceptionTime, which consists of six inception modules. For illustration, we selected 16 hay-harvesting grassland sample points, labeled (1)–(16) in the subplots of Fig. 13. The periods of mowing events occurred in these examples were highlighted by red transparent boxes in Fig. 13(a)–(g). For visual identification, a pair of HLS RGB images before and after each mowing event was also presented for each sample [Fig. 13(h)].

LITE demonstrated high adaptive sensitivity to disturbances caused by mowing events in the NDVI curves. The inception module of LITE exhibited the strongest focus on time steps directly surrounding the mowing events. [Fig. 13(a)]. LITE

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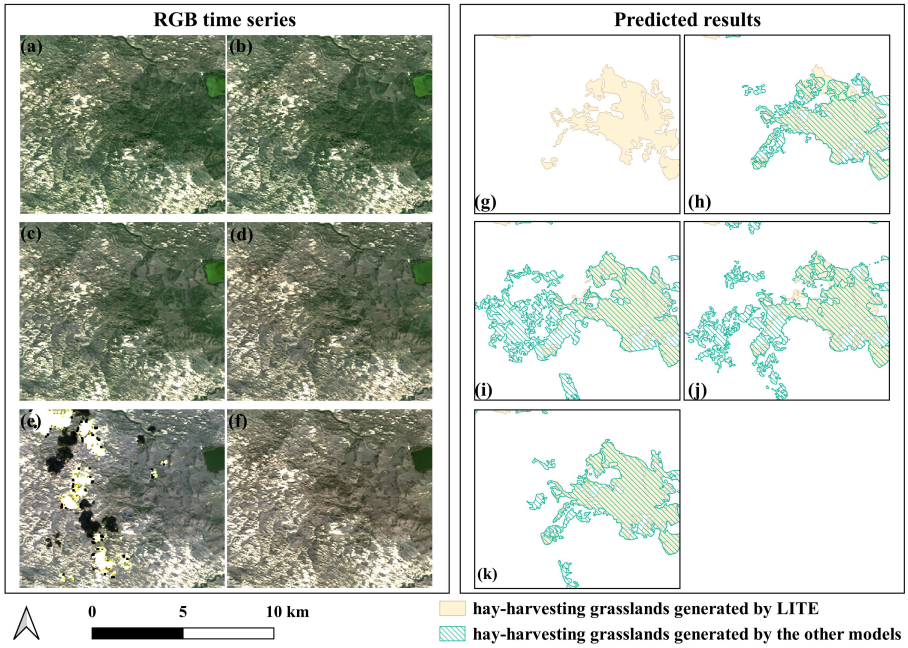


Fig. 10. Comparison of results generated by different models in a selected region neighboring to temperate desert steppe. (a) DOY = 231. (b) DOY = 239. (c) DOY = 247. (d) DOY = 255. (e) DOY = 264. (f) DOY = 271. (g) LITE. (h) LITE versus RF. (i) LITE versus MLP. (j) LITE versus 1D-CNN. (k) LITE versus inception time.

also exhibited focused attention between June 1 (DOY 152) and July 15 (DOY 196), corresponding to the flowering period.

Different inception modules within InceptionTime exhibited varying attention to time steps [Fig. 13(b)–(g)]. For the first inception module within InceptionTime, the most important time steps are difficult to discern, as the attention score peaks appear disordered [Fig. 13(b)]. For the second inception module of InceptionTime [Fig. 13(c)], we observed a consistent attention peak between approximately June 9 (DOY 160) and July 9 (DOY 190), corresponding to the flowering period, regardless of the timing of mowing events. For the third module [Fig. 13(d)], we observed the adaptive attention peak centered around the mowing events. Very little consistent information can be extracted from the fourth module due to the disordered attention score peaks as shown in Fig. 13(e). The fifth Inception module of InceptionTime focused on the period surrounding the mowing events, as well as on the interval from May 20 (DOY 140) to August 3 (DOY 215), encompassing the flowering and flourishing stages [Fig. 13(f)]. The sixth inception module of InceptionTime exhibited broadly distributed attention, with high scores across most time steps [Fig. 13(g)].

Overall, the six inception modules of InceptionTime are prone to capturing redundant and noninformative features from time series data. Merely, the single inception module of LITE can extract information that is highly related to mowing events. The visual comparisons of model interpretation result further highlight the superiority and simplicity of LITE.

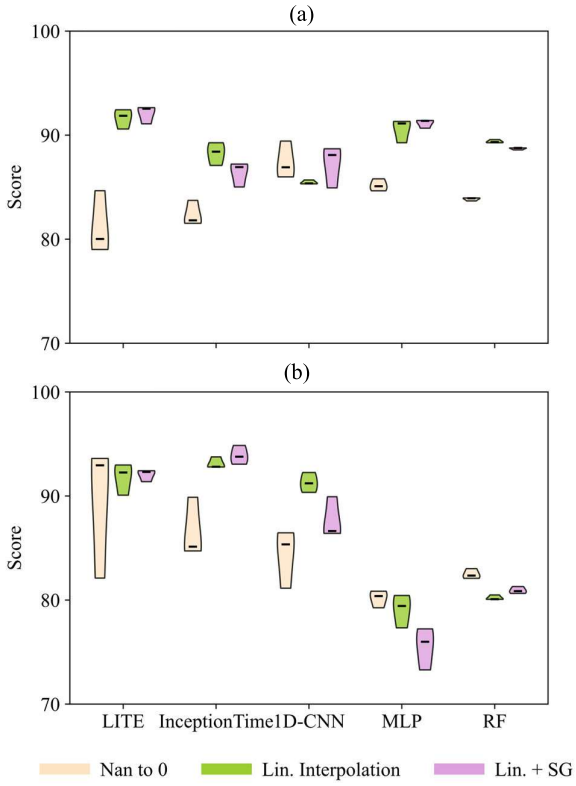


Fig. 11. Results under different gap handling methods. Nan to 0: missing values were filled with zeros; Lin. Interpolation: gaps were filled using a linear interpolation approach; Lin.+SG: linear interpolation was applied followed by SG filtering. The black line in each violin box represents the median value. (a) Precision. (b) Recall.

V. Discussion

A. Difference From Recent Studies

Existing studies [15], [16], [17] were typically conducted within meadow parcels with predefined boundaries, showing dependencies on the availability of field boundaries. The parcel-based approaches may cause false detections when grassland-monitoring activities are spatially or temporally heterogeneous within a parcel [14]. Our pixel-based method is

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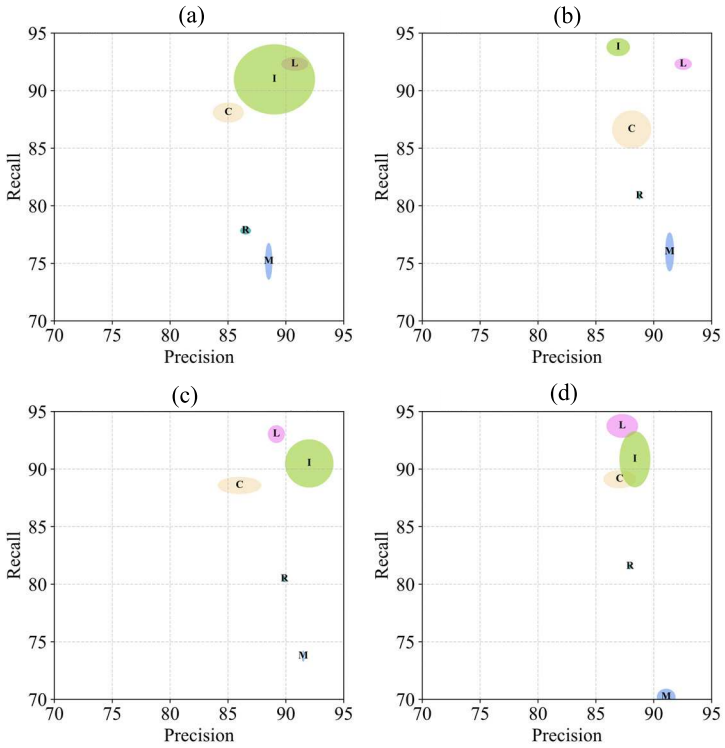


Fig. 12. Bubble plot comparing the recall and precision of different models over different time intervals L: LITE; I: InceptionTime; C: 1D-CNN; M: MLP; R: RF. The coordinates of the ellipse centroids represent median values of recall and precision. The lengths of the two axes of the ellipse denote the standard deviation of the precision and recall, respectively. (a) Time interval of one day(s). (b) Time interval of three day(s). (c) Time interval of five day(s). (d) Time interval of six day(s).

essential for regions where field boundaries are unavailable and management activities are not always homogenous.

Besides, methods in these existing studies, including threshold-based and model-based ones, have proven effective for monitoring meadows characterized by multiple cycles of growth, harvesting, and regrowth in one year. However, when applied to semi-arid regions where grasslands regrow little after mowing [12], [59], these methods predictably demonstrate limited adaptability. Thus, this study pays attention to monitor natural hay-harvesting grasslands in semi-arid temperate steppes. Since prolonged neglect of record-keeping and observation has resulted in a significant lack of data in these regions, our study can contribute to construct valuable geospatial datasets and provide insights for the modernization and management of regional livestock husbandry.

Our quantitative classification result, achieving the highest $F1$-score of 92.06%, is comparable to, or exceeds, those reported in existing studies [14], [17] that employed deep learning models to distinguish mowing from nonmowing time series. Holtgrave et al. [17] trained a LSTM model with NDVI, EVI, and weather time series data, achieving an $F1$-score of 89% with a total dataset of 1200 mowing events. Lobert et al. [14] trained a 1D-CNN model using NDVI time series from Sentinel-2 and SAR features derived from Sentinel-1, achieving an $F1$-score of 83.9% with a total dataset of 257 mowing events. The quantitative comparisons further indicated the effectiveness of our pixel-based method in mapping hay-harvesting grasslands in semi-arid temperate steppes.

B. Model Performance

As shown in Table II, models capable of learning temporal dependencies outperform those treating each time step as an independent feature, highlighting the superiority of time series classification models. Particularly, LITE exhibited the best overall performance among the time-series based models.

As for results under different grassland types showing in Figs. 7 and 8, models generally performed better on temperate meadow steppe than on temperate typical steppe. Given that grassland harvesting is an abrupt disturbance to vegetation growth, postharvest changes in surface biomass are more pronounced in temperate meadow steppe than in temperate typical steppe. Such pronounced changes embedded in input sequence facilitate easier detection by the models. Merely, LITE shows robustness and stability to different grassland types, with precision and recall values exceeding 90% for both temperate typical steppe and temperate meadow steppe. The other models, particularly RF, MLP, and 1D-CNN, resulted in more omissions or errors for different grassland types.

We owe the superiority of LITE over the other models to these aspects.

1) RF and MLP treat each time step as an independent feature and treat the input sequence as a fixed-size vector. RF and MLP process all input time steps simultaneously; therefore, they are unable to capture the temporal dependency characteristics embedded in the input time series [60]. The other models, including 1D-CNN, InceptionTime, and LITE, learn temporal

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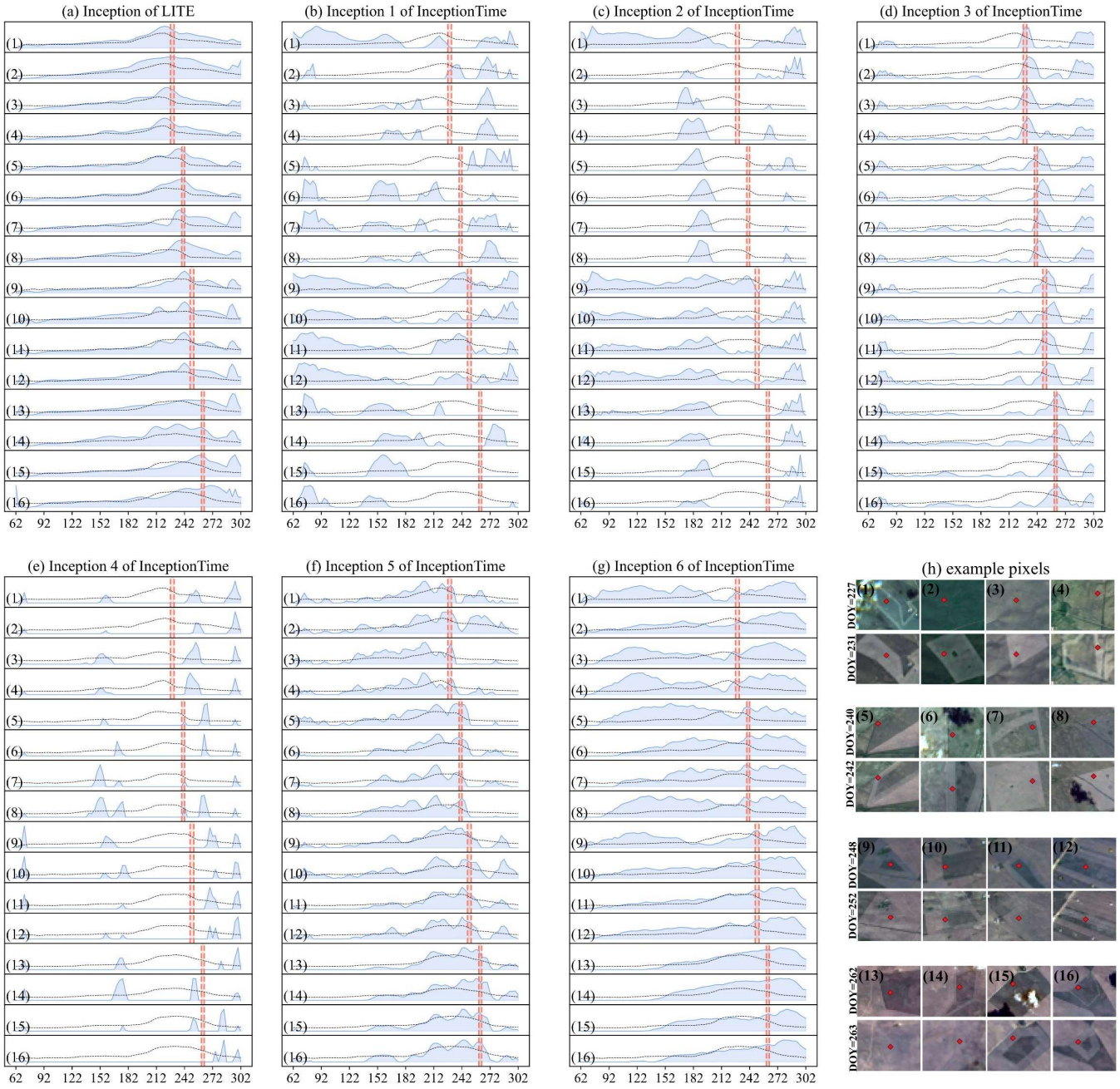


Fig. 13. Time attribution curves derived from the Inception modules of LITE and InceptionTime. x-axis denotes DOY. (1)–(16) represent 16 sample points. Black dotted lines in (a)–(g) represent NDVI curves, while the blue curve envelope areas indicate importance scores of time steps; red transparent boxes indicate the periods when occurred mowing events that can be observed according to (h) at the example pixel locations. (a) Inception of LITE. (b) Inception 1 of InceptionTime. (c) Inception 2 of InceptionTime. (d) Inception 3 of InceptionTime. (e) Inception 4 of InceptionTime. (f) Inception 5 of InceptionTime. (g) Inception 6 of InceptionTime. (h) Example pixels.

information through convolutional layers, which capture temporal dependencies via sliding kernels with continuously updated weights and biases. Therefore, models that capture contextual information from time series data outperform those that disregard temporal order.

2) The 1D-CNN model includes two convolutional layers with a fixed kernel size of 3 in this study, whereas the inception modules in LITE and InceptionTime consist of a bottleneck layer and four convolutional layers with multiscale kernel sizes varying from 1 to 40. The use of multiscale kernel sizes enables LITE and InceptionTime to capture both short- and long-term temporal dependencies [61], [62]. Additionally, the bottleneck layer allows the model to exploit and integrate deep features across different levels. This architectural rigor contributes to the superior performance of LITE and InceptionTime compared to the 1D-CNN.

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3) When comparing LITE and InceptionTime, both of which use inception modules to learn deep features, LITE achieves better results. We attribute the improvement obtained by LITE to its hybrid module, which contains handcrafted convolutional filters to explicitly extract increasing, decreasing, and peak information in time series. Hay-harvesting grasslands grow naturally and prohibit grazing before mowing, which encompasses the rejuvenation, flowering, and flourishing stages. Thus, hay-harvesting grasslands generally exhibit higher growth rates and greater biomass accumulation than nonharvesting grasslands prior to the mowing season. This difference can be directly captured by the increasing filters of LITE. Furthermore, hay-harvested grasslands typically exhibit higher and slightly earlier growth peaks compared to grazed grasslands, due to the absence of livestock trampling and grazing pressure. These variations in growth peak characteristics can be captured by peak filters. Besides, mowing activities cause an abrupt decline in NDVI over a short period, which can be captured by decreasing filters. Thus, these hybrid filters, as auxiliary feature extractors besides the Inception module of LITE, help to extract useful features from the input series. Moreover, the parameters of the hybrid module are nonlearnable, contributing to the lightweight nature of LITE. In contrast, the additional Inception modules in InceptionTime introduce a larger number of learnable parameters, leading to reduced efficiency and increased computational demands. As reported in [32], LITE contains only 2.34% of the number of parameters in InceptionTime. Overall, considering both efficiency and effectiveness, LITE proves to be the most suitable model among those compared for practical applications.

C. Gap Handling

In this study, we utilized HLS data, which integrates Sentinel-2 and Landsat-8, to enhance the availability of remotely sensed observations. However, data gaps are common in optical remote sensing. The influence of gaps may result in the uncertainty of time series NDVI derived from HLS data [63]. To improve the data quality, appropriate gap-handling is essential. Through comparative analysis, we assessed the uncertainty introduced by different gap-handling methods in our time series classification task. According to Fig. 11, for models that effectively capture temporal dependencies, the linear interpolation approach yields notably better performance than the nan-to-zero gap-filling method. That is because zeros may introduce extra noises into the input sequence, impairing model representation ability.

However, the performance differences between using linear interpolation alone and combining it with SG filtering were minimal. Consistent with our findings, Holtgrave at al. [17] reported that the performance improvements resulting from interpolation method and gap-filled method were comparable in their study. We attribute the minimal differences to the outstanding representation ability of convolutional neural networks. After interpolating the input sequence, convolutional operations with learnable parameters can exploit contextual information from neighboring time steps and well detect temporal dependencies embedded in sequences [60]. Besides, the multiscale kernels of the inception module enhanced the ability of learning long-term and short-term information for LITE and InceptionTime. As convolutional kernels slide over time, the values are filtered in accordance with their temporal order, aligning with the objective of SG filter. Therefore, applying the SG filter to the input sequence may lead to performance gains that overlap with those already provided by convolutional neural networks. Future research is encouraged to explore advanced gap-handling methods, including machine learning-based approaches [64], [65], to better understand their effects on relevant applications.

D. Impact of Time Intervals of Series Data

As shown in Fig. 12, LITE demonstrated stability, and robustness, consistently achieving higher $F1$-scores and OA than the other models across various time intervals. Its best performance was observed with a three-day time interval.

Larger time intervals reduce the number of input time steps, thereby improving processing speed. However, longer time intervals increase the risk of missing mowing events that occur between records. Most models showed a decline in $F1$-score and OA as the time interval increased from three to seven days. Nevertheless, LITE exhibited a smaller performance decline than models such as MLP and InceptionTime.

Smaller time intervals provide more complete and continuous information but may reduce computational efficiency and introduce additional challenges in data cube construction. Particularly, when time interval was set to one day, slightly lower accuracies were observed. We attribute the reduction to the gap-filling and filtering applied during preprocessing, which may have smoothed NDVI declines caused by mowing, resulting in classification omissions. The temporal resolution of the source RS data, HLS, is approximately three days. The smoothing effect is more pronounced at a one-day interval than at a three-day interval due to the greater number of data gaps requiring interpolation. Moreover, shorter time intervals may introduce redundancy due to the increased number of time steps in the input data. Overall, a three-day interval is recommended for optimal performance.

E. Limitations, Challenges, and Outlook

One constraint of this study was the lack validation on spatial transferability. The proposed approach demonstrated effectiveness within the study area, which is predominantly characterized by temperate typical steppe with scattered patches of temperate meadow steppe. Nonetheless, the effectiveness of the proposed method under different environments requires further exploration. However, the limited availability of accumulated data, particularly in northern China, will pose a significant challenge for constructing reference dataset. We call for further research to validate our method in other regions.

Another constraint of the proposed method was the lack on the investigation of input feature combinations. Indicators

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such as EVI, leaf area index, normalized difference infrared index, and so on, were also used for detecting mowing events [66]. To improve the generalization of the proposed method, we adopted NDVI, which is widely used and validated as an effective proxy in existing studies [17], as indicator to monitor mowing in this study. The selection of the most adaptive indicators requires further investigation through comprehensive comparative analyses, which will be addressed in future work.

For quality control on time series data, we applied gap-filling to address missing values after removing clouds, shadows, snow, and cloud-adjacent pixels. The linear interpolation and SG filtering approaches were easy to conduct and proven effective for our task. Nonetheless, the interpolated and filtered values, especially during hay-harvesting periods, may have attenuated the signals associated with harvesting, thereby introducing detection omissions [14]. Studies [67], [68], [69] have found that combining SAR and optical data can enhance the quality of time series data.

In addition to direct quality control of the input sequence to enhance performance, nonsatellite datasets can also be incorporated into the task. Since hay harvesting typically begins on predictable sunny days to facilitate natural drying, integrating weather data may aid in more effective information extraction [17]. Merely, the integration of multisource datasets may introduce challenges related to spatial and temporal consistency [70], [71].

This study initially proposed an approach to delineate spatial extents of hay-harvesting grasslands. By addressing the limitation of insufficient spatial information, this study provides a valuable reference for further research and practical applications in semi-arid temperate steppes, thereby supporting the advancement of smart herding and regional sustainable development [72], [73]. Temporal information, such as harvesting dates, is also critical for managing grassland utilization intensity. As shown in Fig. 13, LITE we used primarily focused on periods surrounding mowing events, which suggested that LITE represented specific features directly associated with mowing dates. Therefore, LITE holds potential for mowing date detection, provided that high-quality labeled samples are available. This will be the focus of our future work. Accordingly, with spatial and temporal information, we will conduct further analysis on the impact of climatic factors including soil nutrients, soil moisture, precipitation, and so on, on hay-harvesting management and transferability of detecting methods in future work.

VI. Conclusion

This study initially proposed an approach for mapping hay-harvesting grasslands in temperate steppes. Eurasian semi-arid steppes have been largely overlooked in the construction of geospatial datasets, as existing studies have predominantly focused on meadows in temperate marine climates. Our study offers valuable insights to support the development of geospatial datasets and advance smart herding practices in these underrepresented regions.

The model, LITE, used in this study, achieved an $F1$-score of 92.06% and an OA 90.65% in the study area. LITE's effectiveness and robustness was validated through comparison with RF classifier, MLP, 1D-CNN, and InceptionTime. Furthermore, model interpretation using Grad-CAM revealed that LITE effectively focused on time steps directly associated with harvesting activities. These findings highlight the adaptability and potential of LITE in capturing the temporal dynamics of hay harvesting, which will be the focus of our future research.

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